

Question

Weigh out $\text{M}(\text{NO}_3)_2$ 9.93 g, urea 9.00 g, and PEG – 2000 3 g, dissolve them in 800mL of deionized water, and heat in an oil bath at 105 °C. After reacting for 5h, filter under suction. Wash the product three times each with deionized water and anhydrous ethanol solution, then dry at 80 °C for 8h to obtain 6.709 g of basic salt **A** with a yield of 86.56% (based on **M**). Thermogravimetric analysis of **A** shows an initial weight loss of 2.2%, corresponding to H_2O , followed by a combined weight loss of 11.3% in the second and third steps, accompanied by the release of a colorless and odorless gas, yielding oxide **B**.

By inference, determine which of the following options is correct.

- A.** The square of the period number of element **M** is less than 20
- B.** Basic salt **A** contains more than four elements
- C.** The gas released during the second and third weightlessness periods contains water.
- D.** The gas composition released during the second and third weightlessness periods consists of more than one component.
- E.** The oxidation state of the metal element in oxide **B** differs from that in basic salt **A**.
- F.** In the simplest chemical equation for the reaction generating basic salt **A** in this problem, the sum of the stoichiometric coefficients of all reactants is greater than 20.

Answer

Correct Answer: **F**

Detailed Explanation

Let A contain n M, and B be MO, then:

$$nm(\mathbf{A})/\alpha M(\mathbf{A}) = m(\mathbf{M}(\text{NO}_3)_2)/M(\mathbf{M}(\text{NO}_3)_2)$$

If $\omega_{1-1,2,3}$ represents the mass fraction of n B in A (i.e., 1 – total weight loss), then:

$$nm(\mathbf{A})/\alpha M(\mathbf{A}) = \omega_{1-1,2,3}nm(\mathbf{A})/\omega_{1-1,2,3}\alpha M(\mathbf{A}) = \omega_{1-1,2,3}nm(\mathbf{A})/\alpha M(n\mathbf{MO}) = \omega_{1-1,2,3}m(\mathbf{A})/\alpha M(\mathbf{MO})$$

Solving for $M(\mathbf{M}) = 208.5 \text{ g}\cdot\text{mol}^{-1}$, checking the table and combining with the knowledge of elements, we know that $\mathbf{M}$ is Pb, which is a sixth-period element, and the square of the period number is 36, so A is wrong.

CHECKPOINT

3 PTS

$\mathbf{M}$ is Pb

CHECKPOINT

1 PTS

The square of the period number is 36

A is a basic salt, obviously $n > 1$. Combining the three thermogravimetric losses of water and colorless and odorless gas, as well as urea in the reactants, it is preliminarily speculated that the composition of A is $a\text{Pb}(\text{OH})_2 \cdot (n - a)\text{PbCO}_3$. Based on this, enumerate the possibilities. When n is greater than 2 and less than 10, only when $n=3$, $a=1$ meets the requirements.

Therefore, $a = 1$, $n = 3$, that is, the composition of A is $\text{Pb}_3(\text{OH})_2(\text{CO}_3)_2$, and the verification meets the second and third weight loss data. It contains four elements, so option B is wrong.

CHECKPOINT

2 PTS

The composition of A is $\text{Pb}_3(\text{OH})_2(\text{CO}_3)_2$

CHECKPOINT

1 PTS

A has four elements

The relative molecular mass of A is 775.616. After the first weight loss, it no longer contains hydrogen, so the second and third weight losses cannot contain water, and can only be carbon dioxide, each time losing 1 molecule of carbon dioxide. The weight loss is due to the decomposition of carbonate, there is no redox reaction, and there is no valence change.

CHECKPOINT

1 PTS

The second and third weight losses cannot contain water

CHECKPOINT

1 PTS

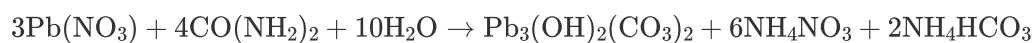
The second and third weight losses are both carbon dioxide

CHECKPOINT

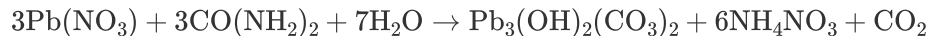
2 PTS

A and **B** have the same valence

The last option requires writing the equation:



or



Therefore, the sum of the stoichiometric numbers of each reactant is greater than 20, which is 26 or 21.

CHECKPOINT

3 PTS

In the chemical reaction equation for the formation of basic salt **A**, the sum of the stoichiometric numbers of each reactant is 26 or 21